

Aarav Verma

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EDUCATION

University of Illinois at Urbana-Champaign

Graduation: May 2029

Bachelor of Science, Aerospace Engineering, Grainger College of Engineering

Relevant Coursework: *Differential Equations, Physics Electricity & Magnetism, CS124: Java, Aerospace Fundamentals*

SKILLS

CAD & Simulation: SolidWorks, Autodesk Fusion, Siemens NX, ANSYS (FEA, Static Structural), OpenRocket, Onshape

Engineering Methods: GD&T, Tolerance Analysis, FEA, Structural Analysis, DFM/DFA, Root Cause Analysis, 6-DOF Dynamics

Programming: Python (PyQt5, Matplotlib, NumPy), MATLAB, Java (OOP, Data Structures), Git, Linux

Professional: Technical Design Reviews, Manufacturing Drawings, BOM Development, Technical Presentations and Collaboration

EXPERIENCE

National Organization for Business and Engineering

Champaign, IL

Engineering Lead, Canary Drone Project

Feb 2026-Present

- Redesigned propeller from standard 2-blade to toroidal configuration in Fusion, reducing blade-tip vortex losses by enclosing the tip path and improving thrust efficiency by ~11% at 4-6k RPM while lowering acoustic signature below 60 dB at 1m hover
- Validated structural integrity of the toroidal propeller and motor mount assembly under 12N thrust loads using FEA in Fusion confirming a factor of safety above 2.0, and developed a fully cohesive BOM tracking 50+ components
- Designed a modular drone airframe in Fusion featuring a Jetson computer housing frame and foldable leg motor assembly with 4 snap-lock hinge joints, reducing field setup time and enabling tool-free reconfiguration between flight missions
- Preparing the Canary drone for the Cozad New Venture Challenge, competing against 300+ teams and 900+ students for \$500,000 in funding by developing investor pitch materials, and refining the final flight ready prototype

Ghost Electric Motorcycles

Champaign, IL

Drivetrain Engineer

Aug 2025-Present

- Engineered a rear suspension swingarm in SolidWorks by reverse-engineering the 2010 Kawasaki ZX-6R, calculating anti-squat ratio and pivot geometry to replicate OEM kinematics, and validating stiffness, braking, and acceleration loads using FEA
- Designed a load-bearing drivetrain frame in SolidWorks to house and align motor, top rod, and swingarm subassemblies ensuring precise mounting interfaces to ± 0.05 mm tolerances
- Designed and prototyped a load-bearing drivetrain top rod in SolidWorks, producing manufacturing-ready drawings with GD&T-defined tolerances and validating fit through iterative 3D-printed prototype testing
- Performed structural analysis on the drivetrain assembly in ANSYS FEA and MATLAB, validating stiffness and load capacity under peak torque deliver, chain tension, and rider-weight loading conditions across breaking, cornering, and max acceleration

Washington State Department of Labor & Industries (L&I)

Olympia, WA

Cybersecurity Analyst Intern

Jun 2024-Jan 2025

- Analyzed 10,000+ daily firewall traffic logs to identify anomalies and implement rule-based mitigation strategies that reduced false-positive alerts by ~30% and improved network uptime
- Automated threat detection and response workflows using Microsoft Defender and PowerShell, reducing manual response time by ~45% through streamlined alert triage

The American Rocketry Challenge (TARC) / Olympia Aerospace Club

Olympia, WA

Founder/Lead Mentor, TARC Team Captain

Sept 2022-Jun 2025

- Designed and optimized high-power rocket airframes and recovery systems in Onshape, validating stability and trajectory using OpenRocket simulations
- Founded and led 10 competitive TARC teams (8th nationally), mentoring 400+ students, securing \$23k+ in sponsorships, and coordinating nationally recognized STEM outreach

PROJECTS

Thrust Vector Control (TVC) Rocket Flight Simulator

Champaign, IL

Python, PyQt5, Matplotlib

Nov 2025-Dec 2025

- Built a 6-DOF rocket flight simulator in Python to develop a foundational understanding of TVC systems incorporating quaternion attitude dynamics, RK4 numerical integration, mass-varying propulsion, and gimbal-to-torque TVC modeling
- Implemented real-time 3D visualization with trajectory trails, velocity vectors, HUD telemetry, and CSV export for flight analysis

Porta Profile

Champaign, IL

Fusion, PostgreSQL, Node.js, React.js

Nov 2025

- Designed a credit-card sized stainless steel NFC multi-tool in Fusion that enables recruiters to instantly transfer contact information to candidates' phones at career fairs, eliminating reliance on paper business cards and lost resumes
- Developed a full-stack web platform (React.js, Node.js, PostgreSQL) with NFC/QR tap-to-share integration, awarded 2nd place at the Illinois Design Challenge

LEADERSHIP & INVOLVEMENT

Illinois Space Society (ISS)

Champaign, IL

Spaceshot, Structural Design & Analysis (SDA) Sub team Engineer

Sept 2025-Dec 2025

- Designed and analyzed Spaceshot structural components in Siemens NX to assess loads and guide design decisions
- Performed ANSYS FEA on fin assemblies to evaluate stress and deformation under launch and flight load